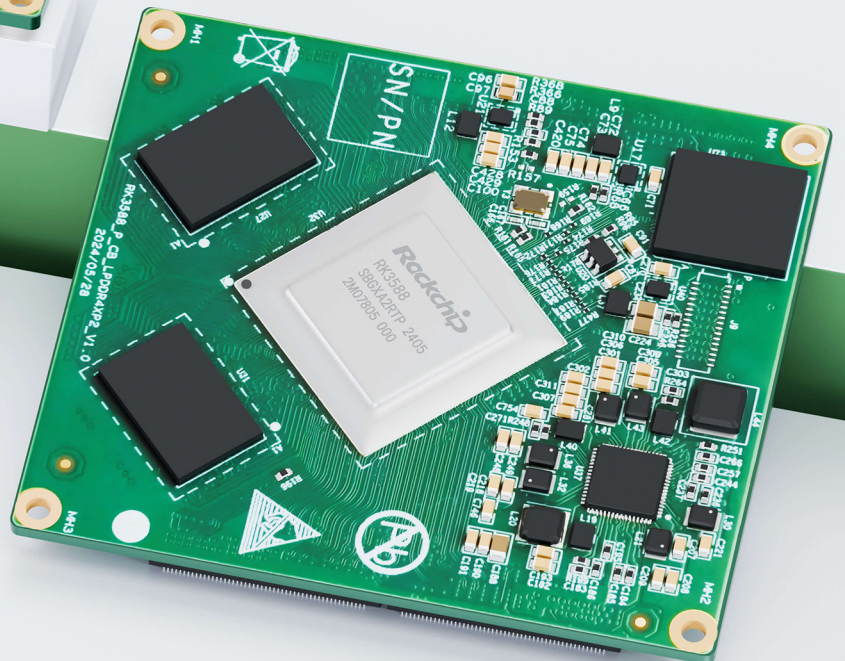
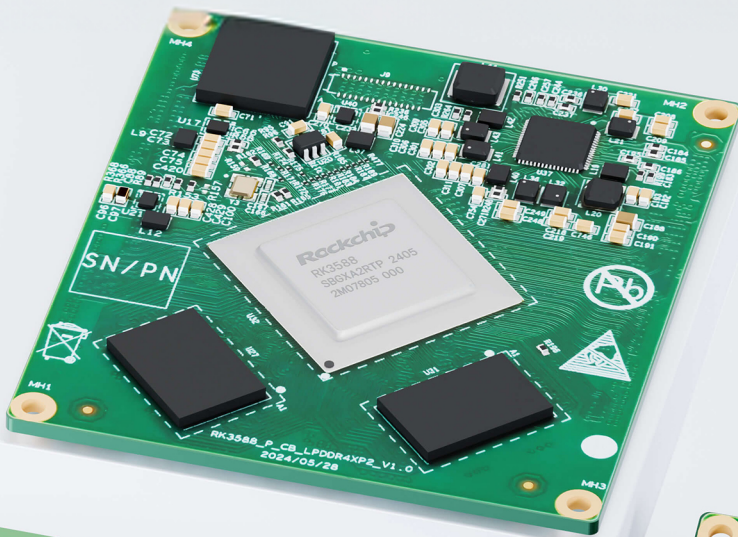


Embrace Edge AI, Empower Industrial Digitalization



EC3588-C AI System on Module

· RK3588 ARM SoC

· 4×A76+4×A55

· 6TOPS NPU

· 8K Video Codec

1. Product Overview

EC3588-C system on module is developed based on the Rockchip RK3588 flagship ARM SoC, delivering powerful performance for high-end embedded applications.

Product Features:

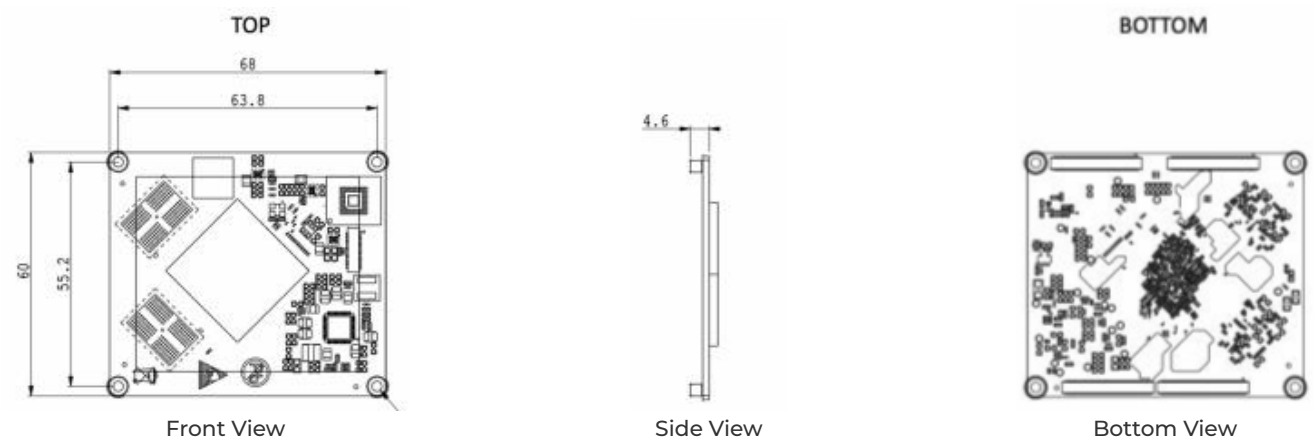
- **High-Performance CPU:** 8nm octa-core, 4×A76+4×A55 @ 2.4 GHz
- **Powerful AI NPU:** 6.0 TOPS triple-core, multi-format acceleration
- **Advanced Graphics:** Mali-G610 MP4, 8K@60fps codec, quad-screen output
- **Rich Interfaces:** PCIe 3.0, USB 3.1, SATA, GMAC, MIPI-CSI/DSI
- **High-End Media:** HDMI RX/TX, DP TX, multi-camera support

Core Technical Specification

Technical Indicator	Specification
OS	Linux
Industrial Protocol	Modbus RTU/TCP, EtherNet/IP, OPC UA, Mitsubishi MC
Remote Management	InHand DeviceLive, HTTPS, SSH
AI	6.0 TOPS NPU
Video Codec	8K@60 fps decode, 8K@30 fps encode
Cloud Platform	Cloud parameter config, container management, firmware management
CPU	4×A76 @ 2.4 GHz + 4×A55 @ 1.8 GHz
Memory / Storage	16 GB LPDDR4 / 64 GB eMMC
Display Interface	HDMI/eDP TX, DP TX, MIPI DSI, quad independent display
Connectivity	2×GMAC, USB 3.1, PCIe 3.0, SATA, MIPI-CSI
Dimensions (W × D × H)	68 × 60 × 4.6 mm

Technical Indicator	Specification
Operating Temperature	0 °C ~ +80 °C

2. Product Dimensions



- Note:
- 1. All dimensions are in millimeters (mm).
 - 2. All dimensions are approximate and for reference only.
 - 3. Dimensioned drawings are not intended for machining.
 - 4. Dimensions are subject to part and manufacturing tolerances.
 - 5. Specifications may change without prior notice.

3. Hardware Specifications

Category/Parameter	Specification
Hardware Platform	
CPU	4 × ARM Cortex-A76 @ 2.4 GHz + 4 × ARM Cortex-A55 @ 1.8 GHz
NPU	6.0 TOPS, triple core, supports int4/int8/int16/FP16/BF16/TF32 acceleration
GPU	ARM Mali-G610 MP4, OpenGL ES 1.1/2.0/3.2, OpenCL 2.2, Vulkan 1.2
VPU	Decode: H.265/VP9 up to 8K@60 fps, H.264 up to 8K@30 fps, AV1 up to 4K@60 fps; Encode: H.265/HEVC, H.264/AVC up to 8K@30 fps

Category/Parameter	Specification
RAM	16 GB LPDDR4
ROM	64 GB eMMC
Video Input	
MIPI DCPHY	2; Supports DPHY or CPHY; 4-lane MIPI DPHY V2.0 (4.5 Gbps/lane); 3-lane MIPI CPHY V1.1 (2.5 Gbps/lane); Supports multiple MIPI-CSI combos
MIPI CSI DPHY	4; 2-lane MIPI DPHY V1.2 (2.5 Gbps/lane); Two 2-lane can be combined to 4-lane
DVP	1; 8/10/12/16-bit standard DVP up to 150 MHz; Supports BT.601/BT.656/BT.1120
HDMI RX	1; Supports HDMI 2.0 (3.4 ~ 6 Gbps) and HDMI 1.4b (250Mbps ~ 3.4 Gbps); HDCP2.3 and HDCP1.4
Video Output	
HDMI/eDP TX	≤2; HDMI up to 7680 × 4320@60 Hz (3/6/8/10/12 Gbps), HDCP2.3; eDP up to 4K@60 Hz (1.62/2.7/5.4 Gbps), HDCP1.3
DP TX	2; DP TX 1.4a, available for USB3.1 Gen 1, supports 1/2/4 lanes; Up to 8192 × 4320@30 Hz; HDCP2.3, HDCP1.3
MIPI DSI	2; MIPI DPHY 2.0 or CPHY 1.1, up to 4K@60 Hz; Dual MIPI-DSI left/right mode, RGB/YUV up to 10-bit
BT.1120	1; RGB up to 8-bit up to 150 MHz; Up to 1920 × 1080@60 Hz
Audio	
I2S	≤4; I2S0/I2S1: 8 lanes, TX/RX, 16-32 bits, up to 192 kHz; I2S2/I2S3: 2 lanes, TX/RX, 16~32 bits, up to 192 kHz
SPDIF	2; 2× 16-bit data storing; Biphasic stereo output
PDM	2; Up to 8 channels, 16~24 bits, up to 192 kHz; PDM primary receive mode
DSM PWM	1; Convert PCM to bitstream digital 1-bit output
Connectivity	

Category/Parameter	Specification
Ethernet	2 × GMAC by RGMII/RMII, 10/100/1000 Mbps
USB 3.1 Gen 1	3; Up to 5 Gbps; 2 × USB3.1 OTG (multiplexed with DP TX, supports USB Type-C and DP Alt); 1 × USB3.1 Host (multiplexed with PIPE PHY2)
USB 2.0 Host	2
PCIe 3.0	2; Supports RC and EP, up to 8 Gbps; Combinations: 1×4, 2×2, 4×1, 1×2+2×1
PCIe 2.0	≤3; Each supports 1 lane, up to 5 Gbps
SATA	≤3; 3 × SATA3.0 controllers, multiplexed by PCIe and USB_HOST2; Supports eSATA up to 6 Gbps
UART	≤10; 2 × 64-bit FIFO for TX/RX; 5/6/7/8-bit serial, up to 4 Mbps; All 10 UARTs support auto flow control
SPI	≤5; Each controller supports two chip select outputs; Master and slave mode configurable
I2C	≤9; 7-bit and 10-bit address modes; Standard mode up to 100 kbps, high-speed mode up to 400 kbps
PWM	≤16; Supports capturing mode
ADC	≤8; 8 × 12-bit single-end input SAR-ADC, up to 1 MS/s
Power	
Input Power	DC 4V
Mechanical	
Dimensions (W × D × H)	68 × 60 × 4.6 mm
Package	Board-to-board connector (4 × 100-pin, 0.4 mm pitch, combined height 1.5 mm)
Mounting Holes	4 × ϕ 2.5 mm
Environment	

Category/Parameter	Specification
Operating Temperature	0 °C ~ +80 °C

4. Software Specifications

Category/Parameter	Specification
Operating System	
OS	Linux
OS Flashing Method	USB OTG
Data Acquisition Protocol (DSA)	
Industrial Protocol	Modbus RTU Master/Slave, Modbus TCP Master/Slave, EtherNet/IP, ISO on TCP, OPC UA Client/Server, Mitsubishi MC 3C/3E/3C OverTCP, Mitsubishi CPU Port, FINS UDP, Host Link, PPI
Electricity Protocol	DLT645-2007, IEC101/104, DNP3.0
Other Protocol	BACnet, CNC
Maintenance and Management	
Upgrade Method	Supports patent upgrade mechanism, local or remote firmware upgrade
Log	Supports local system logs, remote logs, important log power-off preservation
Remote Management	InHand DeviceLive, HTTP, HTTPS, SSH, etc.
DeviceLive Cloud	Supports cloud-based parameter configuration, container management, application and firmware management

5. Ordering Information

Model Code

Model code: EC3588-C

Product Model

Model	CPU	CPU Clock Speed	Operating Temperature
EC3588-C	4 × A76 + 4 × A55	A76 @ 2.4 GHz, A55 @ 1.8 GHz	0 °C ~ +80 °C

6. Contact Us

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